

P8131 Spring 2026 Homework #6 Solution

- Suppose we have the following statistical model: $Y_{ij} = \mu + b_i + e_{ij}$, where $i = 1, \dots, m$ and $j = 1, \dots, n$, $b_i \sim N(0, \sigma_b^2)$, $e_{ij} \sim N(0, \sigma_e^2)$, b_i and e_{ij} are statistically independent for each i and j , and e_{ij} and e_{ik} are statistically independent for any two values $j, k = 1, \dots, n$ and $j \neq k$. Find the variance of Y_{ij} and the covariance and correlation between any two values Y_{ij} and Y_{ik} , $j \neq k$. Which one of the popular covariance patterns does this correspond to?

Solution:

$$Var(Y_{ij}) = Var(b_i) + Var(e_{ij}) = \sigma_b^2 + \sigma_e^2$$

$$Cov(Y_{ij}, Y_{ik}) = Cov(\mu + b_i + e_{ij}, \mu + b_i + e_{ik}) = Var(b_i) = \sigma_b^2$$

$$Corr(Y_{ij}, Y_{ik}) = \frac{Cov(Y_{ij}, Y_{ik})}{\sqrt{Var(Y_{ij})Var(Y_{ik})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2} = \rho$$

$$Var(Y_i) = \begin{pmatrix} \sigma_b^2 + \sigma_e^2 & \sigma_b^2 & \dots & \sigma_b^2 \\ \sigma_b^2 & \sigma_b^2 + \sigma_e^2 & \dots & \sigma_b^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_b^2 & \sigma_b^2 & \dots & \sigma_b^2 + \sigma_e^2 \end{pmatrix} = (\sigma_b^2 + \sigma_e^2) \begin{pmatrix} 1 & \rho & \dots & \rho \\ \rho & 1 & \dots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \dots & 1 \end{pmatrix}$$

This is the compound symmetry covariance.

- A study was conducted involving 27 children, 16 boys and 11 girls. On each child, the distance (mm) from the center of the pituitary to the pterygomaxillary fissure was made at ages 8, 10, 12, and 14. The goal was to study how distance is affected by age and gender. The data file has 5 columns: (1) observation number, (2) child number (1-27), (3) age, (4) distance measurement, and (5) indicator of gender (0 = girl, 1 = boy).

- Make a spaghetti plot to infer how distance is affected by age and gender.
- Assume the model is

$$Y_{ij} = \beta_0 + a_i + b_0 \cdot I_{(sex_i=0)} + b_1 \cdot I_{(sex_i=1)} + \beta_1 \cdot age_{ij} + e_{ij}$$

where $i = 1, \dots, 27$, and $j = 1, \dots, 4$; β_0, β_1 are fixed coefficients; $a_i \sim N(0, \sigma_a^2)$ and $b_k \sim N(0, \sigma_b^2)$ ($k = 0, 1$) are random coefficients, and $e_{ij} \sim N(0, \sigma_e^2)$ is the random error; a_i ($i = 1, \dots, n$), b_k ($k = 0, 1$) and e_{ij} ($i = 1, \dots, n; j = 1, \dots, p$) are mutually independent. Write the model in the marginal form.

- Consider the following marginal model:

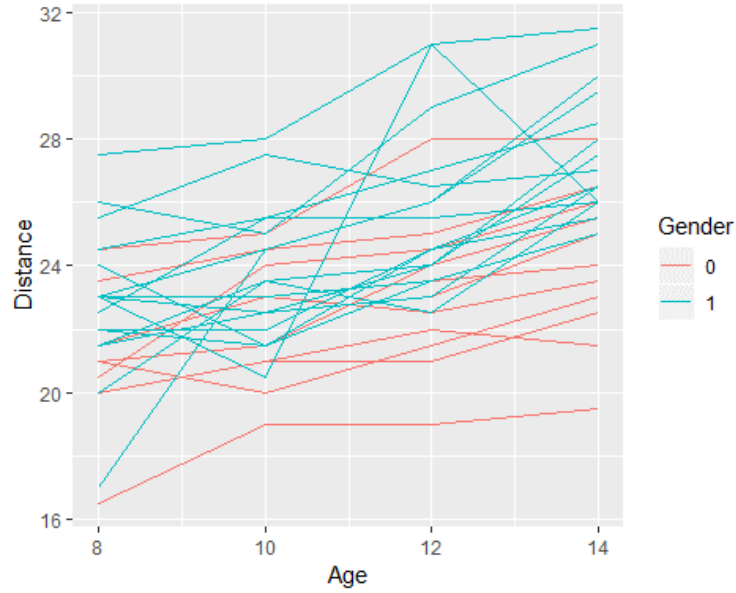
$$E(Y_{ij}) = \beta_0 + \beta_1 \cdot sex_i + \beta_2 \cdot age_{ij}$$

$$Var(Y_i) = \Sigma$$

Fit a model with (a) compound symmetry covariance, (b) exponential covariance, (c) autoregressive covariance. Compare the coefficient parameter estimates and the covariance estimates.

Solution:

- (a) Distance has an increasing trend. Overall boys have larger distances.



- (b) Specify the mean and covariance structure

$$E(Y_{ij}) = \beta_0 + \beta_1 age_{ij}$$

$$Var(Y_{ij}) = Var(a_i) + Var(b_0 * I_{(sex_i=0)}) + Var(b_1 * I_{(sex_i=1)}) + Var(e_{ij}) = \sigma_a^2 + \sigma_b^2 + \sigma_e^2$$

$$Cov(Y_{ij}, Y_{ik}) = \sigma_a^2 + \sigma_b^2$$

$$Var(Y_i) = \begin{pmatrix} \sigma_a^2 + \sigma_b^2 + \sigma_e^2 & \sigma_a^2 + \sigma_b^2 & \dots & \sigma_a^2 + \sigma_b^2 \\ \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 + \sigma_e^2 & \dots & \sigma_a^2 + \sigma_b^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_a^2 + \sigma_b^2 & \sigma_a^2 + \sigma_b^2 & \dots & \sigma_a^2 + \sigma_b^2 + \sigma_e^2 \end{pmatrix} = \Sigma$$

Note that subjects of same sex are correlated, hence $Var(Y)$ (The first 11 subjects being girls, last 16 subjects being boys) would be:

$$\begin{pmatrix} \Sigma & \Sigma^* & \dots & \Sigma^* & 0 & 0 & \dots & 0 \\ \Sigma^* & \Sigma & \dots & \Sigma^* & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma^* & \Sigma^* & \dots & \Sigma & 0 & 0 & \dots & 0 \\ \hline 0 & 0 & \dots & 0 & \Sigma & \Sigma^* & \dots & \Sigma^* \\ 0 & 0 & \dots & 0 & \Sigma^* & \Sigma & \dots & \Sigma^* \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \Sigma^* & \Sigma^* & \dots & \Sigma \end{pmatrix}$$

where

$$\Sigma^* = \sigma_b^2 \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

and 0 is the 4x4 zero matrix.

- (c) We assume that the variance of the response is NOT a constant across age.

The estimated coefficients are

```
##              (Intercept) Gender1 Age
## Compound symmetry      15.40    2.15 0.66
## Exponential            15.35    2.43 0.66
## Autoregressive         15.35    2.43 0.66
```

The estimated compound symmetry covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.613551 3.028079 3.607832 3.340862
## [2,] 3.028079 4.293834 3.155369 2.921880
## [3,] 3.607832 3.155369 6.095417 3.481300
## [4,] 3.340862 2.921880 3.481300 5.226704
```

The estimated exponential covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.742431 3.286644 2.378625 1.325943
## [2,] 3.286644 4.686018 3.391386 1.890498
## [3,] 2.378625 3.391386 6.114274 3.408347
## [4,] 1.325943 1.890498 3.408347 4.733006
```

The estimated autoregressive covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.742471 3.286680 2.378636 1.325964
## [2,] 3.286680 4.686059 3.391395 1.890523
## [3,] 2.378636 3.391395 6.114215 3.408352
## [4,] 1.325964 1.890523 3.408352 4.733036
```

Notice that here the exponential structure gives the exact same results as the autoregressive structure, since we have even time points.

Alternatively, we can assume the variance of the response is a constant across age.

The estimated coefficients are

```
##              (Intercept) Gender1 Age
## Compound symmetry      15.39    2.32 0.66
## Exponential            15.46    2.42 0.65
## Autoregressive         15.46    2.42 0.65
```

The estimated compound symmetry covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.316240 3.266784 3.266784 3.266784
## [2,] 3.266784 5.316240 3.266784 3.266784
## [3,] 3.266784 3.266784 5.316240 3.266784
## [4,] 3.266784 3.266784 3.266784 5.316240
```

The estimated exponential covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.296881 3.315144 2.074839 1.298574
## [2,] 3.315144 5.296881 3.315144 2.074839
## [3,] 2.074839 3.315144 5.296881 3.315144
## [4,] 1.298574 2.074839 3.315144 5.296881
```

The estimated autoregressive covariance matrix is

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 5.296881 3.315144 2.074840 1.298574
## [2,] 3.315144 5.296881 3.315144 2.074840
## [3,] 2.074840 3.315144 5.296881 3.315144
## [4,] 1.298574 2.074840 3.315144 5.296881
```

Appendix

```
library(nlme)
library(ggplot2)
```

```
dental <- read.table("HW6-dental.txt", header=TRUE)
dental$Gender <- as.factor(dental$Gender)
```

```
# 2(a).
p = ggplot(dental, aes(Age, Distance, group=Child, col=Gender)) + geom_line()
print(p)
```

```
# 2(c). We assume that variances of the response are not constant across age.
```

```
# a) compound symmetry covariance
```

```
comsym <- gls(Distance~Gender + Age, dental,
correlation=corCompSymm(form = ~ 1|Child),
weights=varIdent(form = ~ 1| Age), method="REML")
summary(comsym)
#correlation:
temp <- corMatrix(comsym$modelStruct$corStruct)[[1]]
```

```
# ratio between the variances
```

```
ratio <- 1/varWeights(comsym$modelStruct)[1:4]
```

```
# covariance:
```

```
cov_1<-diag(ratio) %*% temp %*% diag(ratio) * comsym$sigma^2
```

```

# b) exponential covariance
expo <- gls(Distance~Gender + Age, dental,
correlation=corExp(form = ~ 1 |Child),
weights=varIdent(form = ~ 1| Age), method="REML")
summary(expo)

#correlation:
temp <- corMatrix(expo$modelStruct$corStruct)[[1]]
# ratio between the variances
ratio <- 1/varWeights(expo$modelStruct)[1:4]
# covariance:
cov_2<-diag(ratio) %*% temp %*% diag(ratio) * expo$sigma^2

# c) autoregressive covariance
auto1 <- gls(Distance~Gender + Age, dental,
correlation=corAR1(form = ~ 1 |Child),
weights=varIdent(form = ~ 1| Age), method="REML")
summary(auto1)
#correlation:
temp <- corMatrix(auto1$modelStruct$corStruct)[[1]]
# ratio between the variances
ratio <- 1/varWeights(auto1$modelStruct)[1:4]
# covariance:
cov_3<-diag(ratio) %*% temp %*% diag(ratio) * auto1$sigma^2
result1 <- rbind(round(comsym$coefficients,2),
round(expo$coefficients,2),
round(auto1$coefficients,2))
rownames(result1) <- c("Compound symmetry", "Exponential", "Autoregressive")

#####
###Alternative:We can assume variance of the response is constant across age. ##
#####
comsym <- gls(Distance~Gender + Age, dental,
correlation=corCompSymm(form = ~ 1|Child), method="REML")
summary(comsym)
# covariance:
cov_1a <- comsym$sigma^2*corMatrix(comsym$modelStruct$corStruct)[[1]]

# b) exponential covariance
expo <- gls(Distance~Gender + Age, dental,
correlation=corExp(form = ~ 1 |Child), method="REML")

```

```

summary(expo)
corMatrix(expo$modelStruct$corStruct)[[1]]
# covariance:
cov_2a <- expo$sigma^2*corMatrix(expo$modelStruct$corStruct)[[1]]

# c) autoregressive covariance
auto1 <- gls(Distance~Gender + Age, dental,
correlation=corAR1(form = ~ 1 |Child), method="REML")
summary(auto1)
corMatrix(auto1$modelStruct$corStruct)[[1]]
# covariance:
cov_3a <- auto1$sigma^2*corMatrix(auto1$modelStruct$corStruct)[[1]]
result2 <- rbind(round(comsym$coefficients,2),
round(expo$coefficients,2),
round(auto1$coefficients,2))
rownames(result1) <- c("Compound symmetry", "Exponential", "Autoregressive")

```